



MINEW BeaconPlus Frame Definition

Version: V005

Date: 3/21/2025

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- Apple iBeacon				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x1A	/
4	1	Manufacture Data	0xFF	/
5	2	Company ID	0x4C00	(little-endian) 0x004C
7	1	Beacon Type	0x02	(big-endian)0x02
8	1	Data Len	0x15	Beacon AdvdataLen
9	16	UUID	0x0112233445566778 899AABBCCDDEEFF0	(big-endian) 0x011223344556677889 9AABBCCDDEEFF0
25	2	Major	0x03E8	(big-endian)0x03E8
27	2	Minor	0x07D0	(big-endian)0x07D0
29	1	RSSI at 1m	0xC5	-59dBm

- Fake iBeacon Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Complete list of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xF1FF	(little-endian) 0xFFFF1
7	1	Data Length	0x17	
8	1	Service data	0x16	/
9	16	UUID	0x0112233445566778 899AABBCCDDEEFF0	(big-endian) 0x011223344556677889 9AABBCCDDEEFF0
25	2	Major	0x03E8	(big-endian)0x03E8
27	2	Minor	0x07D0	(big-endian)0x07D0
29	1	RSSI at 1m	0xC5	-59dBm
30	1	Battery Level	0x64	Battery level is 100%

- Eddystone UID				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completest of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xAAFE	(little-endian) 0xFEAA
7	1	Data Length	0x15	/
8	1	Service data	0x16	/
9	2	UUID data	0xAAFE	(little-endian) 0xFEAA
11	1	Frame Type	0x00	/
12	1	Ranging Data	0x00	Calibrated txpwrat 0m
13	10	Namespace	0x0112233445566778 8990	(big-endian) 0x0112233...
23	6	Instance ID	0xaabbccdde00	(big-endian) 0xaabbccdde00

- Eddystone URL				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completest of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xAAFE	(little-endian) 0xFEAA
7	1	Data Length	0x10	0x07 + sizeof(Encoded URL)
8	1	Service data	0x16	/
9	2	UUID data	0xAAFE	(little-endian) 0xFEAA
11	1	Frame Type	0x10	/
12	1	Ranging Data	0x00	Calibrated txpwrat 0m
13	1	URL Scheme Prefix	0x00	http://www.
14	1-16	Encoded URL	"minewtech"	/
15+	1	HTTP URL encoding	0x00	.com/

- Eddystone TLM				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completestlist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xAAFE	(little-endian) 0xFEAA
7	1	Data Length	0x11	/
8	1	Service data	0x16	/
9	2	UUID data	0xAAFE	(little-endian) 0xFEAA
11	1	Frame Type	0x20	/
12	1	Version	0x00	/
13	2	Battery Voltage	0x0BB8	(big-endian) 3000mV
15	2	Temperature	0x1973	(8.8 fixed-point) 25.44°C
17	4	Adv Count	0x000003E8	(big-endian) 1000 times
21	4	Seconds Count	0x00000258	(big-endian)60 senconds

- MINEW HT Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x10	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x01	/
13	1	Battery level	0x64	Battery level is 100%
14	2	Temperature	0x1973	(8.8 fixed-point) 25.44°C
16	2	Humidity	0x4864	(8.8 fixed-point) 72.39%
18	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- MINEW TEMP Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x0E	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x13	/
13	1	Battery level	0x64	Battery level is 100%
14	2	Temperature	0x1973	(8.8 fixed-point) 25.44°C
16	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- Minew ACC-Axis Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x12	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x03	/
13	1	Battery level	0x64	Battery level is 100%
14	2	X-axis	0x0000	(8.8 fixed-point) 0.00g
16	2	Y-axis	0xFF80	(8.8 fixed-point) -0.50g
18	2	Z-axis	0x013B	(8.8 fixed-point) 1.23g
20	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- MINEW Light lux Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x0E	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x05	/
13	1	Battery level	0x64	Battery level is 100%
14	2	Lux Data	0x4000	16384lux
16	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- MINEW Pressure Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x0E	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x10	/
13	1	Battery level	0x64	Battery level is 100%
14	2	pressure Data	0x4000	16384gram
16	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- MINEW PIR Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x0E	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x11	/
13	1	Battery level	0x64	Battery level is 100%
14	2	PIR Data	0x0001	0x0000:The sensor did not detect infrared 0x0001:Sensor detected infrared
16	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- MINEW TVOC Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x0E	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x12	/
13	1	Battery level	0x64	Battery level is 100%
14	2	TVOC Data	0x4000	16384ppb
16	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- MINEW DFU Advertising Payload				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Complete list of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x0B	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x07	/
13	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- Minew Device Info				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completest of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	/	0x0C + sizeof(name)
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x08	/
13	1	Battery level	0x64	Battery level is 100%
14	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00
20	n	Name	"YLWL" (default)	/

- Configuration Advertising				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xAAFE	(little-endian) 0xFFAA
7	1	Data Length	0x11	/
8	1	Completelist of 128-bit Service UUIDs	0x07	/
9	16	UUID data	0x9ECADC240EE5A9E093F304820100287 F	(little-endian)

- Minew Accelerometer and Gyroscope Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x17	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x15	/
13	2	Acc X-axis	0x0000	(8.8 fixed-point) 0.00g
15	2	Acc Y-axis	0xFF80	(8.8 fixed-point) -0.50g
17	2	Acc Z-axis	0x013B	(8.8 fixed-point) 1.23g
19	2	Deg X-axis	0x0000	BE,unit:10mdps,(int16_t) 0x0000=0*10mdps
21	2	Deg Y-axis	0x0044	BE,unit:10mdps,(int16_t) 0x0044=68*10mdps
23	2	Deg Z-axis	0x9E58	BE,unit:10mdps,(int16_t) 0x9E58=-2500*10mdps
25	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- MINEW Digital Pressure Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Complete list of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x10	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x17	/
13	1	Battery level	0x64	Battery level is 100%
14	4	pressure Data	0x04840000	(16.16 fixed-point) 1156.0 hPa
18	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- Minew 3-axis Magnetometer Data				
Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x12	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x16	/
13	1	Battery level	0x64	Battery level is 100%
14	2	X-axis	0x0000	(8.8 fixed-point) 0.00(10mG)
16	2	Y-axis	0x3380	(8.8 fixed-point) 51.50(10mG)
18	2	Z-axis	0x013B	(8.8 fixed-point) 1.23(10mG)
20	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- Minew Vibration Data

Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x11	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x18	/
13	1	Battery level	0x64	Battery level is 100%
14	4	Timestamp	0x12345678	/
18	1	Vibration status	0x00: No Vibration 0x01: In Vibration	/
19	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- **Minew Photoresistance Data**

Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x0E	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x19	/
13	1	Battery level	0x64	Battery level is 100%
14	2	Lumens	1000	/
16	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00

- Minew Tamper Proof Data

Offset	Length	Type	Data	Details
0	1	Data Length	0x02	/
1	1	Flag data type	0x01	/
2	1	Flag data	0x06	/
3	1	Data Length	0x03	/
4	1	Completelist of 16-bit Service UUIDs	0x03	/
5	2	UUID data	0xE1FF	(little-endian) 0xFFE1
7	1	Data Length	0x0D	/
8	1	Service data	0x16	/
9	2	UUID data	0xE1FF	(little-endian) 0xFFE1
11	1	Frame Type	0xA1	0xA1
12	1	Version Number	0x20	/
13	1	Battery level	0x64	Battery level is 100%
14	1	Tamper Proof Status	0x01	0x01: demolished 0x00:normal
15	6	Mac address	0x009078563412	(little-endian) 12:34:56:78:90:00